# .HTTP Status Codes & Server Basics

## Q-1. Use your browser's developer tools to visit any page on Myntra and list the HTTP status code and two key response headers returned for the main document request.

**Ans:**

I opened the Myntra Men T-Shirts page and used Chrome Developer Tools → Network → Doc. I selected the main document request.

- **Website:** Myntra
- **Request Type:** Document
- **HTTP Status Code:** 200 OK
- **Header 1:** `Content-Type: text/html; charset=UTF-8`
- **Header 2:** `Cache-Control: max-age=...`

The 200 OK status code means the webpage was succ

The 200 OK status code means the webpage was successfully requested and delivered by the server.

*Note: Check the actual Cache-Control value in DevTools and enter the value shown for your request.*

## Q-2. Create a simple HTML file and set up a 301 redirect to another page using an .htaccess file or server configuration on localhost.

**Ans:**

**oldpage.html**

```
<!DOCTYPE html>
<html>
<head>
   <title>Old Page</title>
</head>
<body>
```

```
    <h1>This is the Old Page</h1>
</body>
</html>
```

**newpage.html**

```
<!DOCTYPE html>
<html>
<head>
    <title>New Page</title>
</head>
<body>
    <h1>This is the New Page</h1>
</body>
</html>
```

Create a file named `.htaccess` in the same Apache directory and add:

```
Redirect 301 /oldpage.html /newpage.html
```

Now open:

`http://localhost/oldpage.html`

The browser should redirect to:

`http://localhost/newpage.html`

**Explanation:**

A 301 redirect means that a page has been permanently moved to another URL. It helps search engines understand that the old URL should be replaced by the new URL.

## Q-3. Simulate a redirect loop by creating two HTML files that each redirect to the other using meta refresh tags, then describe what happens in your browser and which HTTP status code (if any) you observe.

**Ans:**

**page1.html**

```
<!DOCTYPE html>
<html>
```

```
<head>
   <meta http-equiv="refresh" content="0; url=page2.html">
   <title>Page 1</title>
</head>
<body>
   <h1>Page 1</h1>
</body>
</html>
```

**page2.html**

```
<!DOCTYPE html>
<html>
<head>
   <meta http-equiv="refresh" content="0; url=page1.html">
   <title>Page 2</title>
</head>
<body>
   <h1>Page 2</h1>
</body>
</html>
```

**What happens?**

When I open `page1.html`, it redirects to `page2.html`. Then `page2.html` redirects back to `page1.html`. This continues repeatedly and creates a redirect loop.

The browser may eventually stop loading the pages and show an error such as “Too many redirects” or “This page isn't working.”

**HTTP status code:**

A meta refresh is not an HTTP redirect. The HTML pages themselves can still return **200 OK**, because the browser performs the navigation after receiving the HTML document. No 301 or 302 status code is necessarily returned for the meta-refresh navigation.

**SEO impact:** Redirect loops prevent users and search engines from reaching the final page and can cause crawling and indexing problems.

## Q-4. Pick any broken link from a real website (for example, a non-existent product on Flipkart) and note the exact HTTP status code shown. Explain the difference between 404 and 410 status codes in your own words.

**Ans:**

For a broken or non-existent webpage, the commonly returned status is:

**HTTP Status Code: 404 Not Found**

To verify it, open a deliberately non-existent URL on a real website and check F12 → Network → document → Status Code.

**Difference between 404 and 410:**

- **404 Not Found:** The requested page cannot currently be found. It may be missing, deleted, or the URL may be incorrect.
- **410 Gone:** The page has been permanently removed, and the server specifically indicates that the resource is no longer available.

In simple words:

- **404:** “I can't find this page.”
- **410:** “This page is gone permanently.”

For SEO, both indicate that the requested URL does not provide a usable page, but 410 gives a stronger signal that the content was intentionally and permanently removed.

# Q-5. Sign up for a free Cloudflare account and add any personal or test domain (or use Cloudflare's playground if you don't have a domain). List three features Cloudflare provides for improving website performance or security.

**Ans:**

**Three important Cloudflare features:**

**1. CDN (Content Delivery Network)**

Cloudflare caches and delivers website content from its global network, helping reduce latency and improve page-loading performance.

**2. DDoS Protection**

Cloudflare detects and mitigates Distributed Denial-of-Service attacks, helping keep websites available during malicious traffic spikes.

**3. SSL/TLS / HTTPS**

Cloudflare provides SSL certificates that encrypt traffic between visitors and the website, helping protect data from interception and tampering.